

The cards below represent three different lines of investigation. The third one is more of a giveaway, you should probably consider carefully whether and when to give it out.

Changing Diagrams a Little 1:

Can you add a fence segment, without changing the number of posts?

Changing Diagrams a Little 2:

Can you add a fence segment, without changing the number of safe regions?

Changing Diagrams a Little 3:

Can you add a fence segment while keeping both the number of posts and the number of safe regions the same?

Changing Diagrams a Little 4:

What happens to the counts if you only add a few posts to the fences, without changing their shape?

My Diagram Has Exactly Four Posts (Part 1):

Start with 4 posts. What is the smallest number of fence segments it must have to be connected, and how many safe regions does this smallest diagram have?

My Diagram Has Exactly 4 Posts (Part 2):

Start with 4 posts. How many segments will I need for at least one safe region? What about two safe regions?

Posts Plus Regions (Part 1):

Add a fourth column for the total number of posts and safe regions. Fill in that column for all the examples you made. For example, if an example has 3 posts and 4 safe regions, in the added column you should write $3 + 4 = 7$.

Posts Plus Regions (Part 2):

What can you say about the link between this total and the number of segments?